Ex. No. 07	APPLICATION STATE MANAGEMENT AND SERIALIZATION
Date of Exercise	
Git Hub Host Link	

Aim

To manage application state and serialize/deserialize data using JSON in a Flutter application.

Description

This experiment demonstrates application state management using `setState()` and JSON serialization/deserialization using a `Student` model. Student data is converted from a Dart object into JSON format using `toJson()` and converted back into a Dart object using `fromJson()`. A list of students is displayed dynamically using `ListView.builder()`.

Procedure:

1. A new Flutter project named `state_app` was created using the Flutter command.
2. A Dart file named `student.dart` was created inside the `lib` folder to define the `Student` model.
3. The `Student` class was created with the properties `id`, `name`, and `course`.
4. The `toJson()` method was implemented to convert the Student object into a JSON-compatible `Map`.
5. The `fromJson()` factory constructor was implemented to convert JSON data back into a Student object.
6. A Student object was created with the values 1, Ragu, and CSE.
7. The `toJson()` method was used to serialize the Student object into JSON data.
8. The `fromJson()` method was used to deserialize the JSON data back into a Student object.
9. A `List<String>` was created to maintain the student names as application state.
10. The `setState()` method was used to add a student to the list whenever the Add Student button was pressed.
11. `ListView.builder()` was used to dynamically display the student names.
12. The application was executed and tested in the Chrome browser.

Program

student.dart

```
class Student
 { final int id;
 final String name;
 final String course;
 Student({
  required this.id,
  required this.name,
  required this.course,
 });
 Map<String, dynamic> toJson()
  { return {
   'id': id,
   'name': name,
   'course': course,
  };
 }
 factory Student.fromJson(Map<String, dynamic> json)
  { return Student(
   id: json['id'],
   name: json['name'],
   course: json['course'],
```

```
    );
  }
}
```

main.dart

```
import 'package:flutter/material.dart';
import 'student.dart';

void main() {
  runApp(const StateApp());
}

class StateApp extends StatelessWidget
  { const StateApp({super.key});

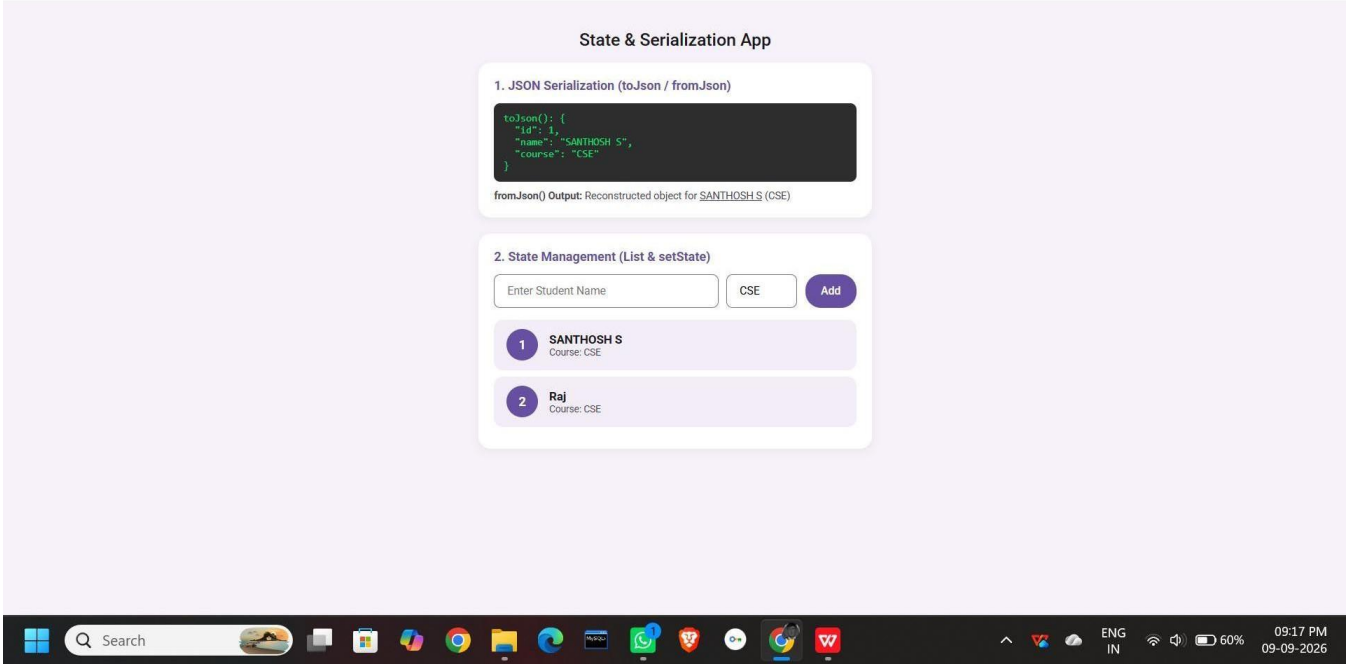
  @override
  Widget build(BuildContext context)
    { return
    MaterialApp(  debugShowCheckedMode
    Banner: false, title: 'State Management',
    home: const StudentPage(),
    );
  }
}
```

```
class StudentPage extends StatefulWidget
  { const StudentPage({super.key});
  @override
  State<StudentPage> createState() => _StudentPageState();
}
class _StudentPageState extends State<StudentPage>
  { List<String> students = [];
  Student student =
    Student( id: 1,
    name: 'Raj',
    course: 'CSE',
  );
  @override
  void initState()
    { super.initState();
    Map<String, dynamic> data = student.toJson();
    print(data);
    Student newStudent = Student.fromJson(data);
    print(newStudent.name);
  }
  void addStudent()
    { setState(()
    { students.add('Raj');
    });
  }
```

```
@override
Widget build(BuildContext context)
 { return Scaffold(
   appBar: AppBar(
     title: const Text('Student State'),
   ),
   body:
     Column( chil
     dren: [
       const SizedBox(height: 20),
       ElevatedButton(
       onPressed: addStudent,
         child: const Text('Add Student'),
       ),
       Expanded(
         child:
           ListView.builder( itemCount:
           students.length, itemBuilder:
           (context, index) { return ListTile(
               title: Text(students[index]),
             );
           },
         ),
       ),
     ],
```

),

Output


State & Serialization App
1. JSON Serialization (toJson / fromJson)
toJson(): {
"id": 1,
"name": "SANTHOSH S",
"course": "CSE"
}
fromJson() Output: Reconstructed object for SANTHOSH S (CSE)
2. State Management (List & setState)
Enter Student Name
CSE
Add
1
SANTHOSH S
Course: CSE
2
Raj
Course: CSE
Search
ENG
IN
60%
09:17 PM
09-09-2026

Result

Thus, the Flutter application was successfully developed to manage changing student data using `setState()`. The Student object was successfully serialized into JSON and deserialized back into a Dart object. The student data was also dynamically displayed using `ListView.builder()`.